- *Infrastructure and platform operations*—In environments that are not cloud native, these are usually called *system administrators* or *sysadmins*. They own the platform (IaaS or containers) and provide (self-service) access to the platform to developers and DevOps roles. They are usually interested in low-level signals, such as utilization of resources and cost-related signals.
- *Nonengineering roles*—A product owner is often called a *product (or project) manager* and owns a feature, a service, or an entire application, depending on the scope and size. These roles usually focus on end user–facing indicators as well as performance and usability. Another nontechnical role is that of a business stakeholder. They represent the part of the organization that specifies the requirements and usually focuses on financial indicators, such as usage, revenue, or units sold. For this role, technical indicators are usually irrelevant. Table 1.1 shows example observability flows across the roles discussed in this section.

Table 1.1 Example flows per role

Role	Example flow
Developer	Uses traces to figure out which microservice along the request path is slow
Developer	Uses continuous profiling to speed up the hot path of a microservice
DevOps	Receives an alert about a component being down and then uses metrics plotted in a dashboard to understand the alert in context
DevOps	Uses metrics to narrow down components causing high latency; then jumps to traces to drill into a specific request path; and then looks at logs of one microservice, serving on that request path for details (correlation)
SRE	Tracks goals and their fulfillment, advising developers via dashboards
Platform operator	Uses a cluster-level dashboard to track autoscaling over time
Release engineer	Uses a dashboard to monitor the build progress of a microservice artifact (e.g., a container image) to, for example, prevent performance regressions
Product manager	Uses a CD dashboard to monitor deployment progress of microservices
Business stakeholder	Uses a dashboard to track weekly revenue and SLA violations

Let's now have a look at a concrete example of a cloud-native system, the signal sources, and telemetry.

1.4 Example microservices app

We will be using Weaveworks's microservices application, Sock Shop (https://microservices-demo.github.io/), to explore observability. Figure 1.2 shows its overall architecture. The Sock Shop example app is a simple demo app built using different languages and frameworks (Spring Boot, Go kit, and Node.js) and is packaged using Open Container Initiative–compliant container images. This makes it widely usable from container orchestrators, such as Kubernetes, Docker Compose,